

QUADRATIC EXPRESSION

- **Polynomial:** - Any expression $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$, where $a_0, a_1, a_2, \dots, a_n$ are constants and $a_n \neq 0$, is called as a polynomial of degree n .
- **Equation:** - An expression of equality is known as equation or an expression equals to another expression is known as equation.
- **Algebraic Equation:** - If any equation involves polynomial in one variable, then it is called as algebraic equation.
- **Transcendental Equation:** - If any equation involves trigonometric, logarithmic, exponential expression, then it is called as transcendental equation.
- **Different Types of Algebraic Equation:** -
 1. **Linear Equation:** - An equation which contains linear polynomial is called as linear equation.
Example: - $ax + b = 0$
 2. **Quadratic Equation:** - An equation which contains quadratic polynomial is called as quadratic equation.
Example: - $ax^2 + bx + c = 0$
 3. **Cubic Equation:** - An equation which contains cubic polynomial is called as cubic equation.
Example: - $ax^3 + bx^2 + cx + d = 0$
 4. **Biquadratic Equation:** - An equation which contains biquadratic polynomial is called as biquadratic equation.
Example: - $ax^4 + bx^3 + cx^2 + dx + e = 0$
- **Solution of Algebraic Equation:** - For any value of x which satisfies $f(x) = 0$ are called as solution of $f(x) = 0$ or root of $f(x) = 0$.
Example: - For $2x + 6 = 0$
 - a. If $x = 2$, it is not satisfies $2x + 6 = 0$. So, $x = 2$ is not a solution of $2x + 6 = 0$.
 - b. If $x = -3$, it is not satisfies $2x + 6 = 0$. So, $x = -3$ is a solution of $2x + 6 = 0$.
- **General Properties of roots of Equation:** -
 1. If $\sim$ is a root of the equation $f(x) = 0$, then $f(x)$ is exactly divisible by $x - \sim$ or $x - \sim$ is a factor of $f(x)$.
 2. If $f(x)$ is exactly divisible by $x - \sim$, then $\sim$ is a root of $f(x) = 0$.
 3. Every equation of n th degree has exactly n roots, may be real, and may be imaginary.
 4. Let $f(x) = 0$ such that, $f(a)$ and $f(b)$ have different sign then equation $f(x) = 0$ has at least one root lies between a and b .

5. In an equation with real coefficients, if $\alpha + i\beta$ is a root then $\alpha - i\beta$ is also one of the root.
Or imaginary roots always occur in conjugate pairs.
6. If $f(x) = 0$ has $\alpha + \sqrt{\beta}$ is one of the root, then $\alpha - \sqrt{\beta}$ is also another root.
7. Every equation of odd degree has atleast one real root.

• **Descarte's Rule of Sign:-**

- a. Maximum number of positive real root of $f(x) = 0$ is same as the number of sign changes in $f(x)$.
 - b. Maximum number of negative real root of $f(x) = 0$ is same as the number of sign changes in $f(-x)$.
- If an equation having degree n such that at most p number of positive roots, at most q number of negative roots, then equation has atleast $n - (p + q)$ number of imaginary roots.

- **Quadratic Equation:** An equation of the form $ax^2 + bx + c = 0$ where $a \neq 0, b, c$ are coefficients is called as quadratic equation.

- The roots of the quadratic equation $ax^2 + bx + c = 0$ are given by $x = \frac{-b \pm \sqrt{D}}{2a}$ where $D = b^2 - 4ac$ is called as discriminant of quadratic equation.

- The quadratic equation whose two roots α, β are given is $\Rightarrow x^2 - (\text{sum of roots})x + \text{product of roots} = 0 \Rightarrow x^2 - (\alpha + \beta)x + \alpha\beta = 0$.

- If α, β are the roots of the equation $ax^2 + bx + c = 0$ then:

1. Sum of roots $= \alpha + \beta = -\frac{b}{a} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$.

2. Product of roots $= \alpha\beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}$.

3. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(-\frac{b}{a}\right)^2 - 2\frac{c}{a} = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}$.

4. $|\alpha - \beta| = \sqrt{(\alpha + \beta)^2 - 4\alpha\beta} = \sqrt{\left(-\frac{b}{a}\right)^2 - 4\frac{c}{a}} = \sqrt{\frac{b^2}{a^2} - \frac{4c}{a}} = \sqrt{\frac{b^2 - 4ac}{a^2}} = \frac{\sqrt{b^2 - 4ac}}{|a|}$.

- In the quadratic equation $ax^2 + bx + c = 0$,
 1. If a and c are of same sign then, product of roots $\frac{c}{a}$ is positive.
 2. If a and c are of opposite sign then, product of roots $\frac{c}{a}$ is negative.
 3. If $a + b + c = 0$ then roots are 1 and $\frac{c}{a}$.
 4. If $a - b + c = 0$ then roots are -1 and $-\frac{c}{a}$.
 5. If the roots are negative, then a, b, c will have same sign.
 6. If the roots are positive, then a, c will have same sign different from sign of b .
- The roots of $ax^2 + bx + a = 0$ are reciprocal to each other.
- **Nature of roots:** In the quadratic equation $ax^2 + bx + c = 0$, coefficients a, b, c are rationals:
 1. If $D = b^2 - 4ac > 0$ then roots are real and distinct.
 2. If $D = b^2 - 4ac > 0$ and is a perfect square then roots are rational and unequal.
 3. If $D = b^2 - 4ac > 0$ and is a not perfect square then roots are irrational and unequal.
 4. If $D = b^2 - 4ac = 0$ then roots are rational and equal.
 5. If $D = b^2 - 4ac < 0$ then roots are imaginary and conjugate to each other.
 6. If $ax^2 + bx + c = 0$ has more than two roots, then this is an identity i.e. $a = b = c$.
- **Nature of the Graph of $f(x) = ax^2 + bx + c$, $a \neq 0, b, c \in R$:**
 - (1) The shape of the curve is a parabola.
 - (2) Y-axis is the axis of parabola.
 - (3) If $a > 0$, the parabola opens upwards.
 - (4) If $a < 0$, the parabola opens downwards.
 - (5) If $D = b^2 - 4ac > 0$, the parabola cuts x-axis at two points.
 - (6) If $D = b^2 - 4ac = 0$, the parabola touch x-axis at one points and the co-ordinate of vertex of parabola is $\left(\frac{-b}{2a}, \frac{-D}{4a}\right)$.
 - (7) Length of latus rectum is $\frac{1}{|a|}$.
 - (8) If $D = b^2 - 4ac < 0$, the parabola does not cut x-axis at any points.
- **Greatest & Least Value of Quadratic Expression:**
 - (1) If $a > 0$, then the expression $f(x) = ax^2 + bx + c$, $a \neq 0, b, c \in R$ has no greatest value but it has least value $\frac{4ac - b^2}{4a}$ at $x = \frac{-b}{2a}$.

- (2) If $a < 0$, then the expression $f(x) = ax^2 + bx + c$, $a \neq 0, b, c \in R$ has no least value but it has greatest value $\frac{4ac - b^2}{4a}$ at $x = \frac{-b}{2a}$.

- **Sign of Quadratic Expression:** Let α, β are the roots of quadratic expression $f(x) = ax^2 + bx + c = 0$. So at $x = \alpha, x = \beta$ the value of expression $ax^2 + bx + c$ is zero and for other values of x the value of expression $ax^2 + bx + c$ is positive or negative as per the following:

- (1) If α, β are the real and distinct roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D > 0$), then $ax^2 + bx + c > 0$ if $x < \alpha$ or $x > \beta$.
- (2) If α, β are the real and distinct roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D > 0$), then $ax^2 + bx + c < 0$ if $\alpha < x < \beta$.
- (3) If α, β are the real and equal roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D = 0$), then $ax^2 + bx + c \geq 0$ if $a > 0$.
- (4) If α, β are the real and equal roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D = 0$), then $ax^2 + bx + c \leq 0$ if $a < 0$.
- (5) If α, β are the imaginary roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D < 0$), then $ax^2 + bx + c > 0$ if $a > 0$.
- (6) If α, β are the imaginary roots of $f(x) = ax^2 + bx + c = 0$. i.e. ($D < 0$), then $ax^2 + bx + c < 0$ if $a < 0$.

- **Position of roots of a quadratic equation:** Let $f(x) = ax^2 + bx + c$, $a \neq 0, b, c \in R$ be a quadratic expression.

- (1) **Conditions for Both roots to be more than a Real number k :**

- (a) $D > 0$.
- (b) $k < \frac{-b}{2a}$.
- (c) $af(k) > 0$.

- (2) **Conditions for Both roots to be less than a Real number k :**

- (a) $D > 0$.
- (b) $k > \frac{-b}{2a}$.
- (c) $af(k) > 0$.

(3) **Conditions for a Real number k lies between Both roots:**

(a) $af(k) < 0$.

(4) **Conditions for exactly one root to lie in between the interval (k_1, k_2) :**

(a) If $f(k_1)f(k_2) < 0$

(5) **Conditions for both the root to lie in between the interval (k_1, k_2) :**

(a) If $a > 0$, $D > 0$, $k_1 < \frac{-b}{2a} < k_2$ and $f(k_1) > 0$ & $f(k_2) > 0$.

(b) If $a < 0$, $D > 0$, $k_1 < \frac{-b}{2a} < k_2$ and $f(k_1) < 0$ & $f(k_2) < 0$

- **Conditions for a quadratic equation to have a repeated roots:** The quadratic equation $f(x) = ax^2 + bx + c = 0$, $a \neq 0, b, c \in R$ has $\sim$ as a repeated roots if and only if $f(\sim) = 0$ and $f'(\sim) = 0$ and the repeated roots are $\sim = \frac{-b}{2a}$.

- **Condition for Two quadratic equations to have a common roots:** Two quadratic equation $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ have a common root if $(a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1) = (c_1a_2 - c_2a_1)^2$.

- **Condition for Two quadratic equations to have equal roots:** Two quadratic equation $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ have equal root if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

- **Note point:**

- (1) To find the common roots of the two quadratic equations, make the coefficient of second degree terms in two equations equal and subtract. The values of x so obtained are the required common root.
- (2) If two quadratic equations with real coefficients have an imaginary roots common, then both roots will be common and the two equations will be identical with $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.
- (3) If two quadratic equations with rational coefficients have an irrational roots common, then both roots will be common and the two equations will be identical with $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

(4) If $\sim$ is a repeated roots of quadratic equation $f(x) = ax^2 + bx + c = 0$, $a \neq 0, b, c \in R$, then $\sim$ is a root of $f'(x) = 0$.

• **Higher Degree Equation:**

• **Relation between Roots and Coefficients of an equation:** -

(1) For the quadratic equation $a_0x^2 + a_1x + a_2 = 0$. If $\sim_1, \sim_2$ are two roots then

$$\sum \sim_i = \sim_1 + \sim_2 = -\frac{a_1}{a_0}$$

$$\sim_1 \cdot \sim_2 = \frac{a_2}{a_0}$$

(2) For the cubic equation $a_0x^3 + a_1x^2 + a_2x + a_3 = 0$. If $\sim_1, \sim_2, \sim_3$ are three roots then

$$\sum \sim_i = \sim_1 + \sim_2 + \sim_3 = -\frac{a_1}{a_0}$$

$$\sum \sim_i \sim_j = \sim_1 \cdot \sim_2 + \sim_2 \cdot \sim_3 + \sim_3 \cdot \sim_1 = \frac{a_2}{a_0}$$

$$\sim_1 \cdot \sim_2 \cdot \sim_3 = -\frac{a_3}{a_0}$$

(3) Similarly, Let us consider nth degree equation $a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n = 0$.

Let $\sim_1, \sim_2, \sim_3, \dots, \sim_n$ are n numbers of roots then

$$\sum \sim_i = -\frac{a_1}{a_0}$$

$$\sum \sim_i \sim_j = \frac{a_2}{a_0}$$

$$\sum \sim_i \sim_j \sim_k = -\frac{a_3}{a_0} \dots \dots \dots$$

$$\sim_1 \cdot \sim_2 \cdot \sim_3 \dots \dots \sim_n = (-1)^n \frac{a_n}{a_0}$$

• **Transformation of Equations:**

(1) If $f(x) = 0$ is a quadratic equation, then equation whose roots are reciprocal of roots

of $f(x) = 0$ is $f\left(\frac{1}{x}\right) = 0$.

(2) If $f(x) = 0$ is a quadratic equation, then equation whose roots are increased by k than the roots of $f(x) = 0$ is $f(x - k) = 0$.

- (3) If $f(x) = 0$ is a quadratic equation, then equation whose roots are decreased by k than the roots of $f(x) = 0$ is $f(x+k) = 0$.
- (4) If $f(x) = 0$ is a quadratic equation, then equation whose roots are negative of the roots of $f(x) = 0$ is $f(-x) = 0$.
- (5) If $f(x) = 0$ is a quadratic equation, then equation whose roots are k times the roots of $f(x) = 0$ is $f\left(\frac{x}{k}\right) = 0$.
- (6) If $f(x) = 0$ is a quadratic equation, then equation whose roots are squares of the roots of $f(x) = 0$ is $f(\sqrt{x}) = 0$.
- (7) If $f(x) = 0$ is a quadratic equation, then equation whose roots are cubes of the roots of $f(x) = 0$ is $f(\sqrt[3]{x}) = 0$.
- (8) If α, β are the roots of the equation $f(x) = 0$, then equation whose roots are $\alpha + b, \beta + b$ is $f\left(\frac{x-b}{a}\right) = 0$.

• **Note Points:**

- (1) Condition for the roots of $ax^2 + bx + c = 0$ are in the ratio $m:n$ is $(m+n)^2 ac = mn b^2$.
- (2) Condition for one of the root of $ax^2 + bx + c = 0$ is square of other is $b^3 + a^2 c + ac^2 = 3abc$.
- (3) Condition for one of the root of $ax^2 + bx + c = 0$ is n th power of other is $(ac^n)^{\frac{1}{n+1}} + (a^n c)^{\frac{1}{n+1}} + b = 0$.
- (4) If the difference of the roots of $ax^2 + bx + c = 0$ is same as the difference of the roots of $px^2 + qx + r = 0$, $\frac{b^2 - 4ac}{q^2 - 4pr} = \frac{a^2}{p^2}$.
- (5) In a quadratic equation with rational coefficients irrational roots occurs as conjugate pairs.
- (6) In a quadratic equation with rational coefficients imaginary roots occurs as conjugate pairs.
- (7) The equation $a_1 x^2 + b_1 x + c_1 = 0$ and $a_2 x^2 + b_2 x + c_2 = 0$ where $a_1 b_2 - a_2 b_1 \neq 0$ have a common root if $(c_1 a_2 - c_2 a_1)^2 = (a_1 b_2 - a_2 b_1)(b_1 c_2 - b_2 c_1)$ and the common roots is $\frac{c_1 a_2 - c_2 a_1}{a_1 b_2 - a_2 b_1}$ or $\frac{b_1 c_2 - b_2 c_1}{c_1 a_2 - c_2 a_1}$.
- (8) If the equation $a_1 x^2 + b_1 x + c_1 = 0$ and $a_2 x^2 + b_2 x + c_2 = 0$ have common roots then $a_1 : a_2 = b_1 : b_2 = c_1 : c_2$.

- (9) If $f(x) = \frac{ax^2 + bx + c}{ax^2 - bx + c}$ or $f(x) = \frac{ax^2 - bx + c}{ax^2 + bx + c}$ and $b^2 - 4ac < 0$, then minimum and maximum value of $f(x)$ are given by $f\left(\pm\sqrt{\frac{c}{a}}\right)$.
- (10) If $f(x) = \frac{(x+a)(x+b)}{x+c}$, then minimum value of $f(x)$ is $(\sqrt{a-c} + \sqrt{b-c})^2$ and it occurs at $-c + \sqrt{(a-c)(b-c)}$ where $a > c, b > c$.
- (11) If $f(x) = \frac{(x+a)(x+b)}{x+c}$, then maximum value of $f(x)$ is $(\sqrt{a-c} - \sqrt{b-c})^2$ and it occurs at $-c - \sqrt{(a-c)(b-c)}$ where $a < c, b < c$.
- (12) An expression of the form $ax^2 + 2hxy + by^2$ where $a, b, h \in C$ is called as homogeneous expression of second degree in x, y .
- (13) An expression of the form $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ where $a, b, c, f, g, h \in C$ and atleast one of where a, b, h is not equal to zero is called a general expression of second degree in x, y .
- (14) The necessary and sufficient condition for second degree expression $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ to be expressed as a product of two linear factors is
- $$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \text{ or } \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0, h^2 \geq ab, g^2 \geq ac, f^2 \geq bc.$$
- (15) If $f(x) = 0$ be any equation and a and b are two real numbers such that $f(a)f(b) < 0$, then equation $f(x) = 0$ has atleast one real root or an odd number of real roots (may be repeated) lies between a and b .
- (16) If $f(x) = 0$ be any equation and a and b are two real numbers such that $f(a)f(b) > 0$, then equation $f(x) = 0$ has either no real roots or an even number of real roots (may be repeated) lies between a and b .

• **Solution of In-equation:**

Here $a > 0$

- (1) $|x| < a \Leftrightarrow -a < x < a$.
- (2) $|x| > a \Leftrightarrow -a < x \cup x > a$.
- (3) If $\sim < \quad$, $(x - \sim)(x - \quad) < 0 \Leftrightarrow \sim < x < \quad$.
- (4) If $\sim < \quad$, $(x - \sim)(x - \quad) \leq 0 \Leftrightarrow \sim \leq x \leq \quad$.
- (5) If $\sim < \quad$, $(x - \sim)(x - \quad) > 0 \Leftrightarrow x < \sim \cup x > \quad$.
- (6) If $\sim < \quad$, $(x - \sim)(x - \quad) \geq 0 \Leftrightarrow x \leq \sim \cup x \geq \quad$.

- **(Wavy Curve Method) To find Method of Intervals where the rational function is positive or negative:**

Let $f(x) = \frac{g(x)}{h(x)} = \frac{(x-a_1)^{m_1}(x-a_2)^{m_2}\dots(x-a_k)^{m_k}}{(x-b_1)^{n_1}(x-b_2)^{n_2}\dots(x-b_r)^{n_r}}$ is > 0 , < 0 , ≥ 0 , ≤ 0 such that

$a_i \neq b_j$ $\begin{pmatrix} i=1,2,\dots,k \\ j=1,2,\dots,r \end{pmatrix}$, m_i, n_j $\begin{pmatrix} i=1,2,\dots,k \\ j=1,2,\dots,r \end{pmatrix}$ are natural numbers.

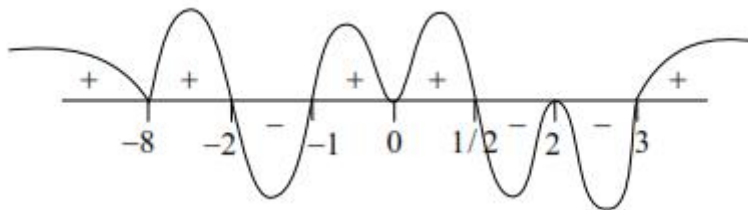
Steps to find sign of the function in the intervals:

- (1) Find all zeroes of $g(x)$ and $h(x)$ and marked in the number line, they are called as critical points.
- (2) All zeroes of $h(x)$ are known as points of discontinuities.
- (3) $g(x)$ and $h(x)$ should not contain common zeroes.
- (4) Even and odd powers of all the zeroes of $g(x)$ and $h(x)$ should be noted.
- (5) Now after marking the numbers on the number line put (+ve) sign in the right of the biggest of these numbers.
- (6) When passing through the next number the polynomial changes its sign if its power is odd and have the same sign if its power is even and continue the process.
- (7) The solution of $f(x) > 0$ is the union of all those intervals in which we have put (+ve) sign.
- (8) The solution of $f(x) < 0$ is the union of all those intervals in which we have put (-ve) sign.

Example: $f(x) = \frac{(x-2)^{442}(x+1)^{123}(x-0.5)^{91}(x+8)^{50}}{x^{92}(x-3)^{87}(x+2)^{41}}$

Solution: Here critical points are -8, -2, -1, 0, 0.5, 2, 3

-8	Even Powers
-2	Odd Powers
-1	Odd Powers
0	Even Powers
0.5	Odd Powers
2	Even Powers
3	Odd Powers



Here $f(x) > 0$ for $(-\infty, -2) \cup (-1, 0.5) \cup (3, \infty)$.

And $f(x) < 0$ for $(-2, -1) \cup (0.5, 3)$.

• **Equation Reducible to Linear, Quadratic and Biquadratic Equations:**

- (1) To solve equation of the type: $(x-a)^4 + (x-b)^4 = A$ put $y = x - \frac{a+b}{2}$.
- (2) To solve equation of the type: $(x-a)^{2n} + (x-b)^{2n} = A$ put $y = x - \frac{a+b}{2}$.
- (3) To solve equation of the type: $a_0(f(x))^{2n} + a_1(f(x))^n + a_2 = 0$ put $(f(x))^n = y$.
- (4) To solve equation of the type: $(ax^2 + bx + c_1)(ax^2 + bx + c_2) \dots (ax^2 + bx + c_n) = A$ put $ax^2 + bx = y$.
- (5) To solve equation of the type: $(x-a)(x-b)(x-c)(x-d) = Ax^2$, where $ab = cd$ put $y = x + \frac{ab}{x}$.
- (6) To solve equation of the type: $(x-a)(x-b)(x-c)(x-d) = A$, where $b-a = d-c$, put $y = x - \frac{a+b+c+d}{4}$ or $y = \frac{(x-a) + (x-b) + (x-c) + (x-d)}{4}$.
- (7) A polynomial $f(x, y)$ is said to be symmetric if $f(x, y) = f(y, x) \forall x, y$.
- (8) Any symmetric polynomial can be represented in the form of $x + y$ or xy .

• **Use of Continuity & Differentiability to find roots of a polynomial:**

Let $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n$, then $f(x)$ is continuous on $\mathbb{R}$.

As $f(x)$ is continuous on $\mathbb{R}$, we can use intermediate value theorem to find whether $f(x)$ has real roots or not. If there exist a and b such that $f(a)f(b) < 0$, then there exist atleast one $c \in (a, b)$ such that $f(c) = 0$. Also if $f(-\infty)$ and $f(a)$ are of opposite sign, then atleast one root lies between $(-\infty, a)$. If $f(a)$ and $f(\infty)$ are of opposite sign, then atleast one root lies between (a, ∞) .

- (1) If $f(x) = 0$ has a root $\sim$ of multiplicity $r (r > 1)$, then we have $f(x) = (x - \sim)^r g(x)$, where $g(\sim) \neq 0$. Also $f'(x) = 0$ has $\sim$ as a root with multiplicity $r - 1$.
- (2) If $f(x) = 0$ has n real roots, then $f'(x) = 0$ has atleast $n - 1$ real roots.
- (3) If $f(x) = 0$ has n distinct real roots, then we can write :

$$f(x) = a_0(x - a_1)(x - a_2) \dots (x - a_n).$$

Then
$$\frac{f'(x)}{f(x)} = \sum_{k=1}^n \frac{1}{x - a_k}.$$

• Cardan's Method Solution of Cubic Equation:

Let us solve $ax^3 + bx^2 + cx + d = 0$ by Cardan's Method

1. Divide equation (1) by coefficient of x^3 i.e. by a, then equation (1) becomes $x^3 + lx^2 + mx + n = 0$ ----- $\rightarrow$ (2)
2. Remove x^2 term from equation (2) by putting $y = x + \frac{l}{3} \Rightarrow x = y - \frac{l}{3}$. Then equation (2) becomes $y^3 + py + q = 0$ ----- $\rightarrow$ (3)
3. Put $y = u + v$ in equation (3), we get
 $\Rightarrow y^3 = u^3 + v^3 + 3uv(u + v) = u^3 + v^3 + 3uvy$
 $\Rightarrow y^3 - 3uvy - (u^3 + v^3) = 0$ ----- $\rightarrow$ (4)
4. From comparing equation (3) and (4) we get
 $uv = \frac{-p}{3}$ and $u^3 + v^3 = -q$
 $\Rightarrow u^3 + v^3 = -q, \quad u^3 v^3 = \frac{-p^3}{27}$
5. u^3, v^3 are the roots of $t^2 - (u^3 + v^3)t + u^3 v^3 = 0$.
 $\Rightarrow t^2 + qt - \frac{p^3}{27} = 0$
 $\Rightarrow t = \frac{1}{2} \left(-q \pm \sqrt{q^2 + \frac{4p^3}{27}} \right)$
 So, $\Rightarrow u^3 = \frac{1}{2} \left(-q + \sqrt{q^2 + \frac{4p^3}{27}} \right), \quad v^3 = \frac{1}{2} \left(-q - \sqrt{q^2 + \frac{4p^3}{27}} \right)$
 Then the values of u are a, a_w, a_w^2
6. As $uv = \frac{-p}{3}$.
 So, corresponding values of v are $\frac{-p}{3a}, \frac{-p}{3a_w}, \frac{-p}{3a_w^2}$.
7. Then find the values of y by putting the values of u and v in $y = u + v$.
8. Then finally find the values of x by putting the values of y in $x = y - \frac{l}{3}$.

• **Ferrari's Method Solution of Biquadratic Equation:**

Let us solve $x^4 + ax^3 + bx^2 + cx + d = 0$ by Ferrari's method.-----→(1)

1. By combining x^4 and x^3 terms into a perfect square equation (1) can be written as

$$\left(x^2 + \frac{a}{2}x + \frac{a^2}{4}\right)^2 - (mx + n)^2 = 0 \text{ .-----}(2)$$

$$\text{Where } 2^a - m^2 + \frac{a^2}{4} = b \quad \Rightarrow m^2 = 2^a + \frac{a^2}{4} - b$$

$$\left. \begin{array}{l} a^a - 2mn = c \quad \Rightarrow 2mn = a^a - c \\ a^2 - n^2 = d \quad \Rightarrow n^2 = a^2 - d \end{array} \right\} \text{-----}\rightarrow(3)$$

2. Eliminate m and n from (3)
i.e. $m^2 \times n^2 = (mn)^2$ and find the value of a .
3. Then put value of a in equation (3) to find m and n.
4. Next put value of a , m and n in (2) and find the values of x.
